

RH AI predictive and generative implementation *Classified - {description}*

Version draft 0.1

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Chapter 1. Project Information

1.1. Document Conventions

1.1.1. Admonitions and Call-Outs



Additional detail about a topic will be called out with the "info" symbol.



Tips and advice will appear with a "lightbulb" admonition.



Warnings and strong recommendations will appear next to a caution exclamation.



Crucial advice and information will be appear next to a red exclamation.

1.1.2. Single-line Commands and Code

If instructed to type a single command, it will be displayed in a monospace font and highlighted like this example:

```
ipa -vvv cert-find --all
```

1.1.3. Multi-line Commands, Code / File Contents

If instructed to type multi-line commands, create or modify code blocks and or file contents the information will be monospaced and placed in a callout box like this example:

```
require 'sinatra'

get '/hi' do
  "Hello World!"
end
```

1.2. Terminology

Understanding and alignment of common vocabulary is critical to any engagement's success. The following table is a reference guide of acronyms/words/phrases and their definitions/meanings that appear in this document.

Table 1. General Terminology

Term	Definition
OCP	OpenShift Container Platform
OCS	OpenShift Container Storage
ODF	OpenShift Data Foundation

Term	Definition
NFS	Network File System
PV	Persistent Volume
PVC	Persistent Volume Claim
RWO	Read Write Once
RWX	Read Write Many
OB	Object Bucket
OBC	Object Bucket Claim
IPI	Instaler Provisioned Infrastructure
UPI	User Provisioned Infrastructure
SDN	Software-Defined Network
RHEL	Red Hat Enterprise Linux
AD	Active Directory
DNS	Domain Name System
DHCP	Dynamic Host Configuration Protocol
FQDN	Fully Qualified Domain Name
VIP	Virtual IP address
VLAN	Virtual LAN is a networking virtualisation technology
VM	Virtual machine
OVS	Open vSwitch
CRD	Custom Resource Definition
LLM training process	* Data Collection and Preparation * Tokenization * Pre-training * Fine-tuning * Evaluation and Testing
Neural Net	* Neurons (nodes) * Layers * Connections * Activation functions * Forward propagation * Backpropagation
LLM	An LLM (Large Language Model) is an advanced artificial intelligence system trained on vast amounts of text data to understand and generate human language. At its core, an LLM exists primarily as a massive collection of weight parameters organized in a neural network architecture.
Upstream	Refers toward source-code origin end of open source software development process and related open source community collaboration, documentation, and testing.
Downstream	Refers toward “productized” deployable, supported artifact (signed binary) end of open source software development process and related contracted errata, event-based support artifacts, product engineering, and product-specific documentation.

1.3. Overview

1.3.1. Executive Summary

Testbed project for knowledge gaining end experiments on the following:

- AI Fundamentals ([Internal](#))
- Red Hat AI Products Overview ([Internal](#))
- Red Hat OpenShift AI Technical Overview ([Internal](#))
- Red Hat Enterprise Linux AI Technical Overview ([Internal](#))
- Introduction to Large Language Models ([Internal](#))
- Practical Approach to RHEL AI Series ([Internal](#))
- Advanced InstructLab Workshop - Train a LLM for the Parasol Insurance Company ([Internal](#))
- Red Hat's Rapid AI Evolution: From InstructLab to Real-World Impact with Shadowbot and Composer AI ([Internal](#))

Chapter 2. Architecture Overview

2.1. RHOAI Overview

1. <https://www.redhat.com/en/blog/how-instructlabs-synthetic-data-generation-enhances-llms>
2. <https://www.youtube.com/watch?v=snMUJGXozec>

Neural Network

A neural network is a computational model inspired by the human brain's structure. Here's a concise description: A neural network consists of:

Neurons (nodes): Basic processing units that receive inputs, apply weights and activation functions, and produce outputs
Layers: Collections of neurons organized into:

Input layer: Receives initial data
Hidden layers: Process information between input and output
Output layer: Produces final results

Connections: Links between neurons with associated weights that determine the influence of one neuron on another
Activation functions: Mathematical operations (like ReLU, sigmoid, or tanh) that introduce non-linearity, allowing networks to learn complex patterns
Forward propagation: The process of data flowing through the network from input to output
Backpropagation: The learning mechanism where errors are calculated at the output and propagated backward to adjust weights

Neural networks learn by iteratively processing examples, comparing outputs to desired results, and adjusting weights to minimize the difference. This ability to learn from data allows them to recognize patterns, make predictions, and perform various tasks across domains like image recognition, natural language processing, and game playing.

LLM

At its core, an LLM exists primarily as a massive collection of weight parameters organized in a neural network architecture. Here's a more specific breakdown: An LLM consists of:

Weight matrices and bias vectors: The billions or trillions of numerical parameters that define the model's behavior. These weights are what the model "learns" during training.
Network architecture: Typically based on the Transformer architecture with attention mechanisms, consisting of multiple layers of neural network components (attention heads, feed-forward networks, etc.). The weights are organized within this architecture in specific mathematical relationships that allow the model to process token sequences and predict next tokens.

When deployed:

The weights and architecture are loaded into memory (often distributed across multiple GPUs)
Input text is tokenized (converted to numerical token IDs)
These tokens flow through the network, interacting with the weights
The output is a probability distribution over the vocabulary

The model isn't a traditional database that stores facts directly. Instead, the "knowledge" is implicitly encoded in the patterns of weights throughout the network. This is why LLMs can produce fluent text but may sometimes generate factually incorrect information - they're reproducing patterns rather than retrieving explicit facts.

LLM training process

The LLM training process involves several key phases:

Data Collection and Preparation: Gathering massive text datasets from the internet, books, articles, and other sources. This data is cleaned, filtered, and formatted appropriately. **Tokenization:** Converting raw text into tokens (words, subwords, or characters) that the model can process. **Pre-training:** The primary training phase where the model learns to predict the next token in a sequence. This is unsupervised learning on the full dataset:

The model processes billions of examples. For each sequence, it predicts what comes next. The difference between predictions and actual tokens creates a loss signal. Weights are updated through backpropagation to reduce this loss.

Fine-tuning: Additional training on more specific datasets, often using:

Supervised fine-tuning (SFT) with human-created examples. Reinforcement learning from human feedback (RLHF) where human preferences guide model outputs.

Evaluation and Testing: Rigorous testing across various tasks to ensure performance, safety, and alignment with human values.

The entire process requires enormous computational resources, often using distributed systems with hundreds or thousands of GPUs running for weeks or months. Each generation of models typically increases in parameter count, training data size, and computational budget.

TensorFlow

TensorFlow is an open-source machine learning framework developed by Google. Here's a short description: TensorFlow is:

- A comprehensive library for building and training machine learning models, especially neural networks
- Built around tensors (multi-dimensional arrays) that flow through computational graphs
- Designed with flexibility to deploy models across different platforms (CPUs, GPUs, TPUs, mobile devices)
- Popular for both research and production applications

Key features include:

- Eager execution for immediate evaluation of operations
- A high-level Keras API for quick model building
- TensorFlow Extended (TFX) for production ML pipelines
- TensorFlow.js for browser-based ML
- TensorFlow Lite for mobile and edge devices
- Extensive visualization tools (TensorBoard) *Support for distributed training across multiple machines

TensorFlow enables developers to create complex neural networks relatively easily while handling the mathematical operations and optimizations behind the scenes. It's used in various applications including computer vision, natural language processing, reinforcement learning, and time series analysis.

Branch and Bound Algorithm:

An algorithmic technique for solving combinatorial optimization problems Works by systematically dividing the solution space into smaller subproblems ("branching") Uses bounds on the optimal solution value to eliminate subproblems without exploring them fully ("bounding") Maintains a best solution found so far and prunes branches that can't improve upon it Commonly used in integer programming, traveling salesman problems, and other NP-hard problems

Relationship to LLMs: Branch and Bound isn't typically a central algorithm in LLM architecture or training. LLMs primarily use:

Transformer architectures with attention mechanisms Gradient-based optimization methods like Adam Backpropagation for weight updates

However, Branch and Bound concepts might appear in LLM applications in these contexts:

In beam search decoding, where multiple candidate sequences are explored (though this is more like breadth-first search than true Branch and Bound) In some specialized prompt optimization techniques In research exploring more efficient inference methods for LLMs In constraint satisfaction problems that LLMs may be used to solve

The core mechanisms of modern LLMs don't directly incorporate Branch and Bound algorithms in their standard architecture or training methodology.

Noam Chomsky and LLM

Noam Chomsky has had a complex and somewhat contradictory relationship with modern LLMs: While Chomsky made foundational contributions to linguistics that indirectly influenced AI development, he has been a prominent critic of statistical language models like LLMs. His relationship to the field includes: Historical Influence:

Chomsky revolutionized linguistics with his theory of generative grammar, suggesting language has deep universal structures His early work on formal language hierarchies (the Chomsky Hierarchy) provided a theoretical framework for computational linguistics His emphasis on language's recursive nature highlighted the complexity that language systems need to handle

Critiques of LLMs:

Chomsky has strongly criticized statistical, data-driven approaches to language modeling He argues LLMs lack true understanding, merely producing statistical approximations of language In a 2023 New York Times essay, he characterized LLMs as "high-tech plagiarism" and "a form of intellectual fraud" He maintains that human language acquisition involves innate structures that LLMs' statistical approaches don't capture

Conceptual Tension:

LLMs represent the triumph of the statistical, data-driven approach to language that Chomsky argued against. The success of LLMs challenges Chomsky's view that statistical models couldn't adequately capture language. However, some researchers suggest the emergence of grammatical patterns in LLMs may ultimately support aspects of Chomsky's theories about universal grammar emerging from simpler principles.

While Chomsky's linguistic theories provided important context for language AI development, modern LLMs evolved largely in opposition to his approach rather than as an implementation of his ideas.

Semantic Web

The Semantic Web is a vision and extension of the World Wide Web that aims to make internet data machine-readable and interconnected in meaningful ways. Here's a concise explanation: The Semantic Web includes:

Structured Data: Information with well-defined meaning that computers can process, rather than just display
Linked Data: Connecting related information across the web through standardized relationships
Ontologies: Formal representations of knowledge domains with defined relationships between concepts
Standard Technologies: Including RDF (Resource Description Framework), OWL (Web Ontology Language), and SPARQL (query language)

Key goals of the Semantic Web:

Enable machines to "understand" web content
Create a global distributed database by connecting data across websites
Allow automated agents to perform complex tasks by navigating interconnected data
Improve search capabilities beyond keyword matching to concept-based understanding

While the complete vision of the Semantic Web hasn't fully materialized as originally conceived by Tim Berners-Lee, many of its technologies have been adopted in various forms, such as in knowledge graphs (like Google's), structured data markup (Schema.org), and linked open data initiatives. Modern AI approaches, including large language models, have since provided alternative solutions to some of the problems the Semantic Web aimed to solve.

The relationship between the Semantic Web and Large Language Models (LLMs) represents an interesting evolution in how we approach machine understanding of information: Contrasting Approaches:

The Semantic Web relies on structured, explicitly defined relationships in data through formats like RDF, OWL, and formal ontologies. LLMs learn implicit relationships in language through statistical patterns in massive datasets without requiring explicit semantic markup.

Complementary Strengths:

Semantic Web provides precision, explicit reasoning, and verifiable data provenance. LLMs offer flexibility, natural language understanding, and the ability to work with unstructured content.

Integration Possibilities:

LLMs can help generate semantic annotations and metadata for web content. Semantic Web data

can provide grounding and factual verification for LLM outputs Knowledge graphs (an implementation of Semantic Web principles) can enhance LLM capabilities by providing structured knowledge

Historical Shift:

The Semantic Web vision required widespread adoption of complex standards and manual annotation LLMs emerged as an alternative approach that can extract meaning from existing unstructured web content Some view LLMs as achieving parts of the Semantic Web's goals through different means

Some researchers are exploring hybrid approaches that leverage the strengths of both paradigms—using the structured, reliable knowledge of semantic technologies while harnessing the flexibility and natural language capabilities of LLMs.

Palantir

Palantir technologies have incorporated elements of both LLMs and Semantic Web principles: Palantir and Semantic Web Technologies:

Palantir's Gotham platform uses ontology-based data models similar to Semantic Web principles Their knowledge graphs represent entities and relationships in structured formats Palantir implements data integration across disparate sources using semantic linking concepts Their systems allow for complex relationship queries across connected data

Palantir and LLM Integration:

Palantir had been incorporating AI/ML capabilities including LLMs They announced the Artificial Intelligence Platform (AIP) which includes LLM integration LLMs enhance their data analysis by extracting insights from unstructured text Natural language interfaces powered by LLMs allow users to query complex data systems

The Hybrid Approach:

Palantir's strength comes from combining structured semantic data representation with AI capabilities Their systems maintain the reliability of explicit data relationships while leveraging AI for pattern recognition This allows for both precise data governance (Semantic Web strength) and flexible analysis of unstructured information (LLM strength)

Palantir's approach exemplifies how enterprise data platforms can benefit from combining elements of both paradigms - using semantic structures for data integrity while employing LLMs to make this complex data more accessible and to derive insights from unstructured information.

Chapter 3. Architecture Design

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RHEL AI

Chapter 4. Installation

4.1. Installing InstructLab

REF: <https://www.redhat.com/en/blog/how-instructlabs-synthetic-data-generation-enhances-llms>
InstructLab Ref: <https://github.com/instructlab/instructlab?tab=readme-ov-file#install-using-pytorch-without-cuda-bindings-and-no-gpu-acceleration>

1. Set Up Your Development Environment

It's a good idea to work in a virtual environment. This ensures that the dependencies for InstructLab won't conflict with other projects you may have:

Create a Virtual Environment:

```
python3 -m venv venv
```

Activate the Virtual Environment:

On Linux/macOS:

```
source venv/bin/activate
```

1. Clone the InstructLab Repository

Cloning the entire repository is advisable because you'll have access to all of the code, configurations, and documentation. Run:

```
git clone https://github.com/instructlab/instructlab.git
```

Then change into the repository directory:

```
cd instructlab
```

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab$ git clone https://github.com/instructlab/instructlab.git
Cloning into 'instructlab'...
remote: Enumerating objects: 18104, done.
remote: Counting objects: 100% (1055/1055), done.
remote: Compressing objects: 100% (492/492), done.
remote: Total 18104 (delta 822), reused 565 (delta 563), pack-reused 17049 (from 3)
Receiving objects: 100% (18104/18104), 15.79 MiB | 31.39 MiB/s, done.
Resolving deltas: 100% (11970/11970), done.
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab$ ls
InstructLab
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab$ cd instructlab
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ ls
CHANGELOG.md  containers  docs  MAINTAINERS.md  notebooks  README.md  requirements-dev.txt  requirements-vllm-cuda.txt  SECURITY.md  tests  TROUBLESHOOTING.md
CODE_OF_CONDUCT.md  CONTRIBUTING  LICENSE  Makefile  pyproject.toml  requirements  requirements.txt  scripts  src  tox.ini
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ python3 -m venv venv
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ source venv/bin/activate
```

1. Install the Required Dependencies

InstructLab's documentation specifies an installation using PyTorch without CUDA bindings (for CPU-only mode). Check the InstructLab README for any additional details, but in general you should:

Install PyTorch (CPU-only):

```
pip install torch torchvision torchaudio --extra-index-url https://download.pytorch.org/whl/cpu
```

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ source venv/bin/activate
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ pip install torch torchvision torchaudio --extra-index-url https://download.pytorch.org/whl/cpu
Looking in indexes: https://pypi.org/simple, https://download.pytorch.org/whl/cpu
Collecting torch
  Downloading https://download.pytorch.org/whl/cpu/torch-2.6.0%2Bcpu-cp313-cp313-linux_x86_64.whl.metadata (26 kB)
Collecting torchvision
  Downloading https://download.pytorch.org/whl/cpu/torchvision-0.21.0%2Bcpu-cp313-cp313-linux_x86_64.whl.metadata (6.1 kB)
Collecting torchaudio
  Downloading https://download.pytorch.org/whl/cpu/torchaudio-2.6.0%2Bcpu-cp313-cp313-linux_x86_64.whl.metadata (6.6 kB)
Collecting filelock (from torch)
  Downloading filelock-3.18.0-py3-none-any.whl.metadata (2.9 kB)
Collecting typing_extensions>=4.10.0 (from torch)
  Downloading typing_extensions-4.13.2-py3-none-any.whl.metadata (3.0 kB)
Collecting networkx (from torch)
  Downloading networkx-3.4.2-py3-none-any.whl.metadata (6.3 kB)
Collecting Jinja2 (from torch)
  Downloading Jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting fsspec (from torch)
  Downloading fsspec-2025.3.2-py3-none-any.whl.metadata (11 kB)
Collecting setuptools (from torch)
  Downloading setuptools-78.1.0-py3-none-any.whl.metadata (6.6 kB)
Collecting sympy<1.13.1 (from torch)
  Downloading https://download.pytorch.org/whl/sympy-1.13.1-py3-none-any.whl (6.2 MB)
  6.2/6.2 MB 27.6 MB/s eta 0:00:00
Collecting mpmath<1.4.0, >=1.1.0 (from sympy==1.13.1->torch)
  Downloading https://download.pytorch.org/whl/mpmath-1.3.0-py3-none-any.whl (536 kB)
  536.2/536.2 kB 39.2 MB/s eta 0:00:00
Collecting numpy (from torchvision)
  Downloading numpy-2.2.4-cp313-cp313-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (62 kB)
Collecting pillow<8.3.0, >=5.3.0 (from torchvision)
  Downloading pillow-11.2.1-cp313-cp313-manylinux_2_28_x86_64.whl.metadata (8.9 kB)
Collecting MarkupSafe>=2.0 (from Jinja2->torch)
  Downloading MarkupSafe-3.0.2-cp313-cp313-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (23 kB)
  Downloading https://download.pytorch.org/whl/cpu/torch-2.6.0%2Bcpu-cp313-cp313-linux_x86_64.whl (178.6 MB)
  178.6/178.6 MB 67.9 MB/s eta 0:00:00
  Downloading https://download.pytorch.org/whl/cpu/torchvision-0.21.0%2Bcpu-cp313-cp313-linux_x86_64.whl (1.8 MB)
  1.8/1.8 MB 52.0 MB/s eta 0:00:00
  Downloading https://download.pytorch.org/whl/cpu/torchaudio-2.6.0%2Bcpu-cp313-cp313-linux_x86_64.whl (1.7 MB)
  1.7/1.7 MB 46.1 MB/s eta 0:00:00
  Downloading pillow-11.2.1-cp313-cp313-manylinux_2_28_x86_64.whl (4.6 MB)
  4.6/4.6 MB 40.1 MB/s eta 0:00:00
  Downloading typing_extensions-4.13.2-py3-none-any.whl (45 kB)
  Downloading filelock-3.18.0-py3-none-any.whl (16 kB)
  Downloading fsspec-2025.3.2-py3-none-any.whl (104 kB)
  Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
  Downloading networkx-3.4.2-py3-none-any.whl (1.7 MB)
  1.7/1.7 MB 47.0 MB/s eta 0:00:00
  Downloading numpy-2.2.4-cp313-cp313-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (16.1 MB)
  16.1/16.1 MB 32.0 MB/s eta 0:00:00
  Downloading setuptools-78.1.0-py3-none-any.whl (1.3 MB)
  1.3/1.3 MB 43.0 MB/s eta 0:00:00
Installing collected packages: mpmath, typing_extensions, sympy, setuptools, pillow, numpy, networkx, MarkupSafe, fsspec, filelock, Jinja2, torch, torchvision, torchaudio
Successfully installed MarkupSafe-3.0.2 filelock-3.18.0 fsspec-2025.3.2 Jinja2-3.1.6 mpmath-1.3.0 networkx-3.4.2 numpy-2.2.4 pillow-11.2.1 setuptools-78.1.0 sympy-1.13.1 torch-2.6.0%2Bcpu torchaudio-2.6.0%2Bcpu torchvision-0.21.0%2Bcpu typing_extensions-4.13.2
[notice] A new release of pip is available: 24.2 -> 25.0.1
[notice] To update, run: pip install --upgrade pip
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

upgrade psp to newest version

```
pip install --upgrade pip
```

```
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ pip install --upgrade pip
Requirement already satisfied: pip in ./venv/lib64/python3.13/site-packages (24.2)
Collecting pip
  Downloading pip-25.0.1-py3-none-any.whl.metadata (3.7 kB)
  Downloading pip-25.0.1-py3-none-any.whl (1.8 MB)
  1.8/1.8 MB 40.6 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.2
    Uninstalling pip-24.2:
      Successfully uninstalled pip-24.2
  Successfully installed pip-25.0.1
```

```
python<version> -m venv --upgrade-deps venv
source venv/bin/activate
pip install instructlab
```

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
~/Projects/Common/AI/InstructLab/instructlab

1b67116cc2e21f4de9
Stored in directory: /home/spopov/.cache/pip/wheels/10/db/97/9408f70cc86bf94637052f806f4e690b4ebacafcb4e10bc1e
Building wheel for numpy (pyproject.toml) ... done
Created wheel for numpy: filename=numpy-1.26.4-cp313-cp313-linux_x86_64.whl size=9183426 sha256=a476772eaa55536df51d6b694fbc42ed7c35e5f6c35e8ae68c88921519bf868
Stored in directory: /home/spopov/.cache/pip/wheels/8b/2d/9f/b6b46373f328e2ef50388915d31ccacbedac929459b5459bf
Building wheel for rouge-score (pyproject.toml) ... done
Created wheel for rouge-score: filename=rouge_score-0.1.2-py3-none-any.whl size=24998 sha256=4a68c9a924ddfd8e60951ce94f8869c9bda906d42b371d7164f82753dcf30f3e
Stored in directory: /home/spopov/.cache/pip/wheels/44/af/da/5ffc433e2786f0b1a9c6f458d5fb8f611d8eb332387f18698f
Building wheel for sentencepiece (pyproject.toml) ... done
Created wheel for sentencepiece: filename=sentencepiece-0.2.0-cp313-cp313-linux_x86_64.whl size=1423981 sha256=9b307cc882f603b7fd017e8fb4bba3696742797137fc3dfd1dcde
718a5125bda
Stored in directory: /home/spopov/.cache/pip/wheels/02/3d/f0/8dbb11925fbd58340b128ae9334ae7028332dT5a5751738f616
Successfully built llama-cpp-python numpy rouge-score sentencepiece
Installing collected packages: wcwidth, sentencepiece, pytz, appdirs, zstandard, xxhash, urllib3, tzdata, typing-inspection, typeguard, tqdm, toml, tenacity, sniffio, smmap, six, shtab, setproctitle, safetensors, ruamel.yaml, rpd-py, regex, PyYAML, python-dotenv, pygments, pydantic-core, pyarrow, psutil, protobuf, propcache, prompt-toolkit, pathspec, packaging, orjson, numpy, multidict, mdurl, jsonpointer, joblib, jiter, instructlab-quantize, idna, hll, fsspec, frozenlist, docstring-parser, distro, diskcache, dill, click, charset-normalizer, certifi, attrs, annotated-types, aiohappyeyeballs, absl-py, yarl, yamllint, uvicorn, sentry-sdk, requests, referencing, python-dateutil, pydantic, nltk, multiprocessing, markdown-it-py, llama-cpp-python, jsonpatch, httpcore, gitdb, gguf, docker-pycreds, click-didyoumean, anyio,aiosignal, starlette, rouge-score, rich, requests-toolbelt, pydantic-yaml, pydantic-settings, pandas, jonschema-specifications, huggingface-hub, httpx, GitPython, aiohttp, wandb, tyro, tokenizers, starlette-context, sse-starlette, openai, langsmith, jonschema, fastapi, accelerate, transformers, langchain-core, datasets, trl, peft, langchain-text-splitters, instructlab
Attempting uninstall: numpy
Found existing installation: numpy 2.2.4
Uninstalling numpy-2.2.4:
Successfully uninstalled numpy-2.2.4
Attempting uninstall: fsspec
Found existing installation: fsspec 2025.3.2
Uninstalling fsspec-2025.3.2:
Successfully uninstalled fsspec-2025.3.2
Successfully installed GitPython-3.1.44 PyYAML-6.0.2 absl-py-2.2.2 accelerate-1.6.0 aiohappyeyeballs-2.6.1 aiohttp-3.11.16 aiosignal-1.3.2 annotated-types-0.7.0 anyio-4.9.0 appdirs-1.4.4 attrs-25.3.0 certifi-2025.1.31 charset-normalizer-3.4.1 click-8.1.8 click-didyoumean-0.3.1 datasets-2.21.0 dill-0.3.8 diskcache-5.6.3 distro-1.9.0 docker-pycreds-0.4.0 docstring-parser-0.16 fastapi-0.115.12 frozenlist-1.5.0 fsspec-2024.6.1 gguf-0.6.0 gitdb-4.0.12 hll-0.14.0 httpcore-1.0.8 httpx-0.28.1 huggingface-hub-0.30.2 idna-3.10 instructlab-0.17.2 instructlab-quantize-0.1.0 jiter-0.9.0 joblib-1.4.2 jsonpatch-1.33 jsonpointer-3.0.0 jonschema-4.23.0 jonschema-specifications-2024.10.1 langchain-core-0.3.51 langchain-text-splitters-0.3.8 langsmith-0.3.31 llama-cpp-python-0.2.75 markdown-it-py-3.0.0 mdurl-0.1.2 multidict-6.4.3 multiprocess-0.70.16 nltk-3.9.1 numpy-1.26.4 openai-1.74.0 orjson-3.10.16 packaging-24.2 pandas-2.2.3 pathspec-0.12.1 peft-0.9.0 prompt-toolkit-3.0.51 propcache-0.3.1 protobuf-4.25.6 psutil-7.0.0 pyarrow-19.0.1 pydantic-2.11.3 pydantic-core-2.33.1 pydantic-settings-2.8.1 pydantic-yaml-1.4.0 pygments-2.19.1 python-dateutil-2.9.0.post0 python-dotenv-1.1.0 pytz-2025.2 referencing-0.36.2 regex-2024.11.6 requests-2.32.3 requests-toolbelt-1.0.0 rich-13.9.4 rouge-score-0.1.2 rpd-py-0.24.0 ruamel.yaml-0.18.10 safetensors-0.5.3 sentencepiece-0.2.0 sentry-sdk-2.26.1 setproctitle-1.3.5 shtab-1.7.2 six-1.17.0 smmap-5.0.2 sniffio-1.3.1 sse-starlette-2.2.1 starlette-0.46.2 starlette-context-0.3.6 tenacity-9.1.2 tokenizers-0.15.2 toml-0.10.2 tqdm-4.67.1 transformers-4.38.2 trl-0.7.11 typeguard-4.4.2 typing-inspection-0.4.0 tyro-0.9.18 tzdata-2025.2 urllib3-2.4.0 uvicorn-0.34.1 wandb-0.16.6 wcwidth-0.2.13 xxhash-3.5.0 yamllint-1.35.1 yarl-1.19.0 zstandard-0.23.0
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

From your venv environment, verify ilab is installed correctly, by running the ilab command.

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
~/Projects/Common/AI/InstructLab/instructlab

.10 safetensors-0.5.3 sentencepiece-0.2.0 sentry-sdk-2.26.1 setproctitle-1.3.5 shtab-1.7.2 six-1.17.0 smmap-5.0.2 sniffio-1.3.1 sse-starlette-2.2.1 starlette-0.46.2 s
tarlette-context-0.3.6 tenacity-9.1.2 tokenizers-0.15.2 toml-0.10.2 tqdm-4.67.1 transformers-4.38.2 trl-0.7.11 typeguard-4.4.2 typing-inspection-0.4.0 tyro-0.9.18 tzd
ata-2025.2 urllib3-2.4.0 uvicorn-0.34.1 wandb-0.16.6 wcwidth-0.2.13 xxhash-3.5.0 yamllint-1.35.1 yarl-1.19.0 zstandard-0.23.0
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ ilab
Usage: ilab [OPTIONS] COMMAND [ARGS]...

NOTE: ilab no longer supports Python 3.9 or Python 3.12 as of version 0.18.
If you are using either of these Python versions, there is no upgrade path
from ilab 0.17 to newer ilab versions. Please switch to Python 3.10 or 3.11
to use the latest ilab. ilab 0.17 will receive no bug fixes or additional
features. You can check which Python version you are using by running
'python --version'

CLI for interacting with InstructLab.

If this is your first time running ilab, it's best to start with 'ilab init'
to create the environment.

Options:
  --config PATH  Path to a configuration file. [default: config.yaml]
  --version      Show the version and exit.
  --help        Show this message and exit.

Commands:
  config  Command Group for Interacting with the Config of InstructLab.
  data    Command Group for Interacting with the Data generated by...
  model   Command Group for Interacting with the Models in InstructLab.
  sysinfo Print system information
  taxonomy Command Group for Interacting with the Taxonomy of InstructLab.

Aliases:
  chat: model chat
  convert: model convert
  diff: taxonomy diff
  download: model download
  generate: data generate
  init: config init
  serve: model serve
  test: model test
  train: model train
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

Check your InstructLab version:

```
ilab --version
```

```

generate: data generate
init: config init
serve: model serve
test: model test
train: model train
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ ilab --version
ilab, version 0.17.2
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$

```

Now, initialize InstructLab configuration:

```
ilab config init
```

```

1 (venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ ilab config init
2
3 NOTE: ilab no longer supports Python 3.9 or Python 3.12 as of version 0.18. If you are using either of these Python versions, there is no upgrade path fro
4 m ilab 0.17 to newer ilab versions.
5 Please switch to Python 3.10 or 3.11 to use the latest ilab. ilab 0.17 will receive no bug fixes or additional features.
6 You can check which Python version you are using by running 'python --version'
7
8 Welcome to InstructLab CLI. This guide will help you to setup your environment.
9 Please provide the following values to initiate the environment [press Enter for defaults]:
10 Path to taxonomy repo [taxonomy]:
11 'taxonomy' seems to not exist or is empty. Should I clone https://github.com/instructlab/taxonomy.git for you? [y/N]: y
12 Cloning https://github.com/instructlab/taxonomy.git...
13 Generating 'config.yaml' in the current directory...
14 Initialization completed successfully, you're ready to start using 'ilab'. Enjoy!
15 (venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$

```

Optional

Install Other Python Dependencies:

The repository likely contains a requirements.txt file. Install these dependencies with:

```
pip install -r requirements.txt
```

This command will install all the libraries needed by InstructLab as defined by its maintainers.

Official install from InstructLab Gitlab

4.1.1. Install Node and NPM

<https://nodejs.org/en/download>

```

# Download and install nvm:
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.3/install.sh | bash

# in lieu of restarting the shell
\ . "$HOME/.nvm/nvm.sh"

# Download and install Node.js:
nvm install 22

# Verify the Node.js version:
node -v # Should print "v22.15.0".
nvm current # Should print "v22.15.0".

```

```
# Verify npm version:
npm -v # Should print "10.9.2".
```

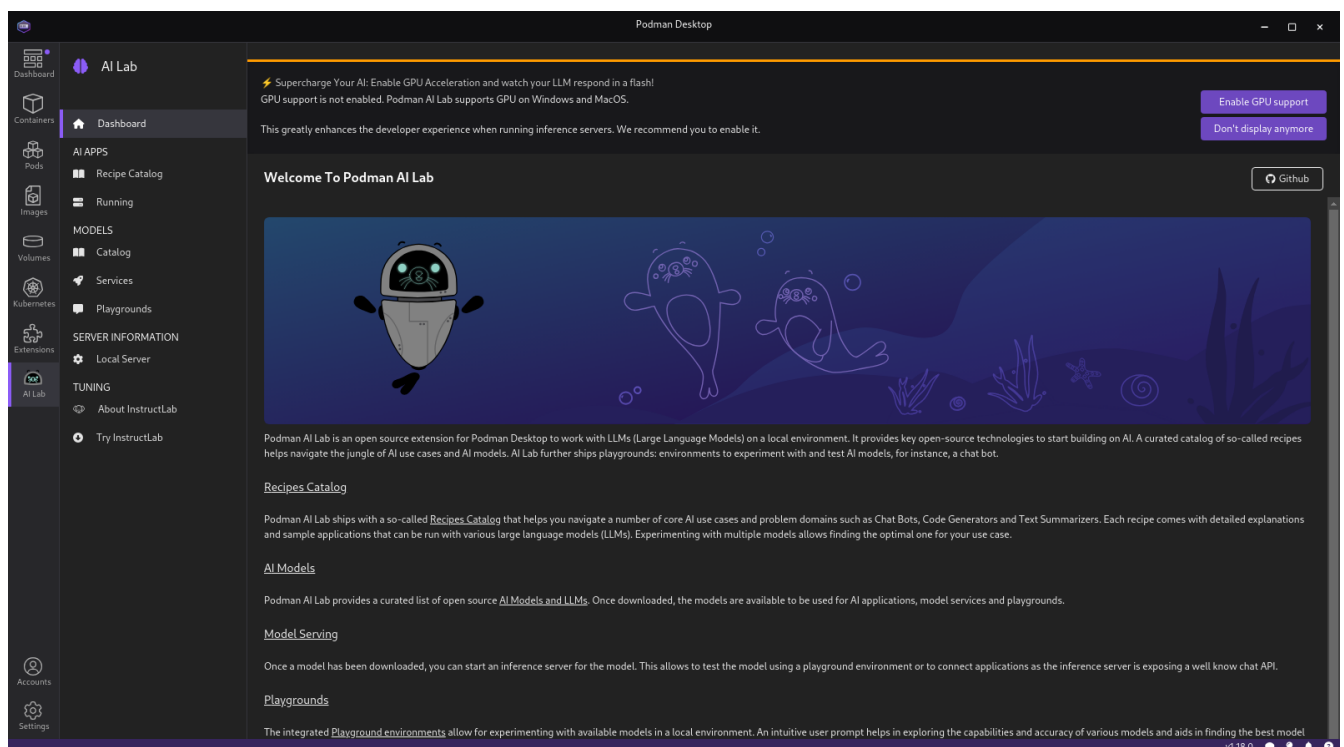
```
spopov@spopov-thinkpadp1gen3:~/Tools/AI/Node
spopov@spopov-thinkpadp1gen3:~/Tools/AI/Node$ curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.3/install.sh | bash
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 16631 100 16631 0 79804 0 --:--:-- --:--:-- --:--:-- 79956
=> Downloading nvm from git to '/home/spopov/.nvm'
=> Cloning into '/home/spopov/.nvm'...
remote: Enumerating objects: 382, done.
remote: Counting objects: 100% (382/382), done.
remote: Compressing objects: 100% (325/325), done.
remote: Total 382 (delta 43), reused 178 (delta 29), pack-reused 0 (from 0)
Receiving objects: 100% (382/382), 385.06 KiB | 3.56 MiB/s, done.
Resolving deltas: 100% (43/43), done.
* (HEAD detached at FETCH_HEAD)
master
=> Compressing and cleaning up git repository

=> Appending nvm source string to '/home/spopov/.bashrc'
=> Appending bash_completion source string to '/home/spopov/.bashrc'
=> Close and reopen your terminal to start using nvm or run the following to use it now:

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh" # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion" # This loads nvm bash_completion
spopov@spopov-thinkpadp1gen3:~/Tools/AI/Node$ export NVM_DIR="$HOME/.nvm"
spopov@spopov-thinkpadp1gen3:~/Tools/AI/Node$ nvm install 22
Downloading and installing node v22.15.0...
Downloading https://nodejs.org/dist/v22.15.0/node-v22.15.0-linux-x64.tar.xz...
##### 100.0%
Computing checksum with sha256sum
Checksums matched!
Now using node v22.15.0 (npm v10.9.2)
Creating default alias: default -> 22 (-> v22.15.0)
spopov@spopov-thinkpadp1gen3:~/Tools/AI/Node$ node -v
v22.15.0
spopov@spopov-thinkpadp1gen3:~/Tools/AI/Node$ nvm current
v22.15.0
spopov@spopov-thinkpadp1gen3:~/Tools/AI/Node$ npm -v
10.9.2
spopov@spopov-thinkpadp1gen3:~/Tools/AI/Node$
```

4.1.2. Install Podman AI Lab extension

Go to Podman extensions and install required for AI Lab



4.1.3. Troubleshooting InstructLab install

1. C compiler is missing

```

Downloading numpy-1.26.4.tar.gz (15.8 MB)
15.8/15.8 MB 44.9 MB/s eta 0:00:00

Installing build dependencies ... done
Getting requirements to build wheel ... done
Installing backend dependencies ... done
Preparing metadata (pyproject.toml) ... error
error: subprocess-exited-with-error

× Preparing metadata (pyproject.toml) did not run successfully.
  exit code: 1
  [20 lines of output]
+ /home/spopov/Projects/Common/AI/InstructLab/instructlab/venv/bin/python3 /tmp/pip-install-56rv858/numpy_278ea489682a4faeba497c860455540d
60455540d /tmp/pip-install-56rv858/numpy_278ea489682a4faeba497c860455540d/.mesonpy-mnscsqqa -Dbuildtype=release -Db_ndebug=if-release -Db_vschr=
nscsqqa/meson-python-native-file.ini
The Meson build system
Version: 1.2.99
Source dir: /tmp/pip-install-56rv858/numpy_278ea489682a4faeba497c860455540d
Build dir: /tmp/pip-install-56rv858/numpy_278ea489682a4faeba497c860455540d/.mesonpy-mnscsqqa
Build type: native build
Project name: NumPy
Project version: 1.26.4

./meson.build:1:0: ERROR: Unknown compiler(s): [['cc'], ['gcc'], ['clang'], ['nvc'], ['pgcc'], ['icc'], ['icx']]
The following exception(s) were encountered:
Running `cc --version` gave "[Errno 2] No such file or directory: 'cc'"
Running `gcc --version` gave "[Errno 2] No such file or directory: 'gcc'"
Running `clang --version` gave "[Errno 2] No such file or directory: 'clang'"
Running `nvc --version` gave "[Errno 2] No such file or directory: 'nvc'"
Running `pgcc --version` gave "[Errno 2] No such file or directory: 'pgcc'"
Running `icc --version` gave "[Errno 2] No such file or directory: 'icc'"
Running `icx --version` gave "[Errno 2] No such file or directory: 'icx'"

A full log can be found at /tmp/pip-install-56rv858/numpy_278ea489682a4faeba497c860455540d/.mesonpy-mnscsqqa/meson-logs/meson-log.txt
[end of output]

note: This error originates from a subprocess, and is likely not a problem with pip.
error: metadata-generation-failed

```

The error you're encountering is coming from the build process of the NumPy package (a dependency) which is being compiled during the installation. The error message indicates that it cannot find any suitable C compilers (like cc, gcc, or clang) on your system. Since NumPy (and some other libraries) often require a C compiler to build from source, you'll need to install one.

This command will install GCC, make, and other necessary tools. You can install GCC specifically with:

```
sudo dnf install gcc gcc-c++ make
```

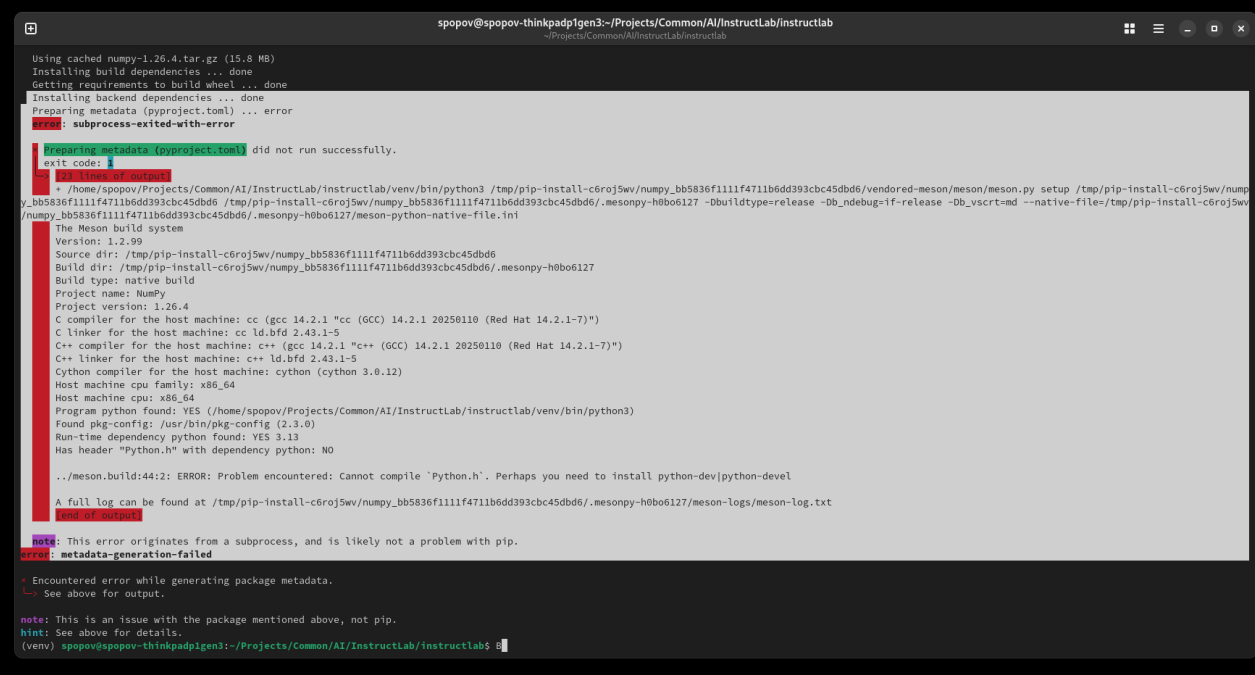
```

spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
~Projects/Common/AI/InstructLab/instructlab
(wenv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ sudo dnf install gcc gcc-c++ make
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository Size
Installing:
gcc x86_64 14.2.1-7.fc41 updates 103.4 MiB
gcc-c++ x86_64 14.2.1-7.fc41 updates 37.7 MiB
make x86_64 1:4.4.1-8.fc41 Fedora 1.6 MiB
Installing dependencies:
cpp x86_64 14.2.1-7.fc41 updates 34.5 MiB
glibc-devel x86_64 2.40-23.fc41 updates 2.3 MiB
kernel-headers x86_64 6.13.3-200.fc41 updates 6.5 MiB
libstdc++-devel x86_64 14.2.1-7.fc41 updates 15.4 MiB
libxcrypt-devel x86_64 4.4.38-7.fc41 updates 30.8 MiB
Transaction Summary:
Installing: 8 packages
Upgrading: 1 package
Replacing: 1 package
Total size of inbound packages is 67 MiB. Need to download 67 MiB.
After this operation, 202 MiB extra will be used (install 202 MiB, remove 289 KiB).
Is this ok [y/N]: y
[1/9] make-1:4.4.1-8.fc41.x86_64 100% | 108.9 KiB/s | 586.1 KiB | 00m05s
[2/9] gcc-0:14.2.1-7.fc41.x86_64 100% | 5.6 MiB/s | 36.1 MiB | 00m00s
[3/9] libstdc++-devel-0:14.2.1-7.fc41.x86_64 100% | 14.5 MiB/s | 2.6 MiB | 00m00s
[4/9] cpp-0:14.2.1-7.fc41.x86_64 100% | 1.7 MiB/s | 11.5 MiB | 00m07s
[5/9] glibc-devel-0:2.40-23.fc41.x86_64 100% | 5.4 MiB/s | 613.8 KiB | 00m00s
[6/9] libxcrypt-devel-0:4.4.38-7.fc41.x86_64 100% | 753.2 KiB/s | 29.4 KiB | 00m00s
[7/9] libxcrypt-0:4.4.38-7.fc41.x86_64 100% | 2.9 MiB/s | 127.5 KiB | 00m00s
[8/9] kernel-headers-0:6.13.3-200.fc41.x86_64 100% | 7.2 MiB/s | 1.6 MiB | 00m00s
[9/9] gcc-c++-0:14.2.1-7.fc41.x86_64 100% | 3.1 MiB/s | 13.7 MiB | 00m04s
-----
[5/9] Total 100% | 6.1 MiB/s | 67.0 MiB | 00m11s
Running transaction
[ 1/12] Verify package files 100% | 21.0 B/s | 9.0 B | 00m00s
[ 2/12] Prepare transaction 100% | 26.0 B/s | 10.0 B | 00m00s
[ 3/12] Upgrading libxcrypt-0:4.4.38-7.fc41.x86_64 100% | 9.8 MiB/s | 291.2 KiB | 00m00s
[ 4/12] Installing kernel-headers-0:6.13.3-200.fc41.x86_64 100% | 16.6 MiB/s | 6.6 MiB | 00m00s
[ 5/12] Installing libxcrypt-devel-0:4.4.38-7.fc41.x86_64 100% | 1.7 MiB/s | 33.1 KiB | 00m00s
[ 6/12] Installing glibc-devel-0:2.40-23.fc41.x86_64 100% | 12.0 MiB/s | 2.3 MiB | 00m00s
[ 7/12] Installing libstdc++-devel-0:14.2.1-7.fc41.x86_64 100% | 39.6 MiB/s | 15.6 MiB | 00m00s
[ 8/12] Installing cpp-0:14.2.1-7.fc41.x86_64 100% | 187.8 MiB/s | 34.6 MiB | 00m00s
[ 9/12] Installing make-1:4.4.1-8.fc41.x86_64 100% | 47.4 MiB/s | 1.8 MiB | 00m00s
[10/12] Installing gcc-0:14.2.1-7.fc41.x86_64 100% | 179.2 MiB/s | 103.4 MiB | 00m01s
[11/12] Installing gcc-c++-0:14.2.1-7.fc41.x86_64 100% | 189.3 MiB/s | 37.7 MiB | 00m00s
[12/12] Removing libxcrypt-0:4.4.38-6.fc41.x86_64 100% | 21.0 B/s | 18.0 B | 00m01s
Complete!
(wenv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ gcc --version
gcc (GCC) 14.2.1 20250110 (Red Hat 14.2.1-7)
Copyright (C) 2024 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

```

```
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ gcc --version
gcc (GCC) 14.2.1 20250110 (Red Hat 14.2.1-7)
Copyright (C) 2024 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

1. Python development package is missing



```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
Using cached numpy-1.26.4.tar.gz (15.8 MB)
Installing build dependencies ... done
Getting requirements to build wheel ... done
Installing backend dependencies ... done
Preparing metadata (pyproject.toml) ... error
error: subprocess-exited-with-error

  Preparing metadata (pyproject.toml) did not run successfully.
  exit code: 1
  19 lines of output
  /home/spopov/Projects/Common/AI/InstructLab/instructlab/venv/bin/python3 /tmp/pip-install-c6roj5wv/numpy_bb5836f111f4711b6dd393cbc45dbd6/vendor-meson/meson/meson.py setup /tmp/pip-install-c6roj5wv/numpy_bb5836f111f4711b6dd393cbc45dbd6 /tmp/pip-install-c6roj5wv/numpy_bb5836f111f4711b6dd393cbc45dbd6/.mesonpy-h8bo6127 -Dbldtype=release -Db_ndebug=if-release -Db_vsct=md --native-file=/tmp/pip-install-c6roj5wv/numpy_bb5836f111f4711b6dd393cbc45dbd6/.mesonpy-h8bo6127/meson-python-native-file.ini
  The Meson build system
  Version: 1.2.99
  Source dir: /tmp/pip-install-c6roj5wv/numpy_bb5836f111f4711b6dd393cbc45dbd6
  Build dir: /tmp/pip-install-c6roj5wv/numpy_bb5836f111f4711b6dd393cbc45dbd6/.mesonpy-h8bo6127
  Build type: native build
  Project name: NumPy
  Project version: 1.26.4
  C compiler for the host machine: cc (gcc (GCC) 14.2.1 20250110 (Red Hat 14.2.1-7))
  C linker for the host machine: ld.bfd 2.43.1-5
  C++ compiler for the host machine: c++ (gcc (GCC) 14.2.1 20250110 (Red Hat 14.2.1-7))
  C++ linker for the host machine: c++ ld.bfd 2.43.1-5
  Cython compiler for the host machine: cython (cython 3.0.12)
  Host machine cpu family: x86_64
  Host machine cpu: x86_64
  Program python found: YES (/home/spopov/Projects/Common/AI/InstructLab/instructlab/venv/bin/python3)
  Found pkg-config: /usr/bin/pkg-config (2.3.0)
  Run-time dependency python found: YES 3.13
  Has header "Python.h" with dependency python: NO
  ../meson.build:44:2: ERROR: Problem encountered: Cannot compile 'Python.h'. Perhaps you need to install python-dev/python-devel
  A full log can be found at /tmp/pip-install-c6roj5wv/numpy_bb5836f111f4711b6dd393cbc45dbd6/.mesonpy-h8bo6127/meson-logs/meson-log.txt
  end of output
note: This error originates from a subprocess, and is likely not a problem with pip.
error: metadata-generation-failed

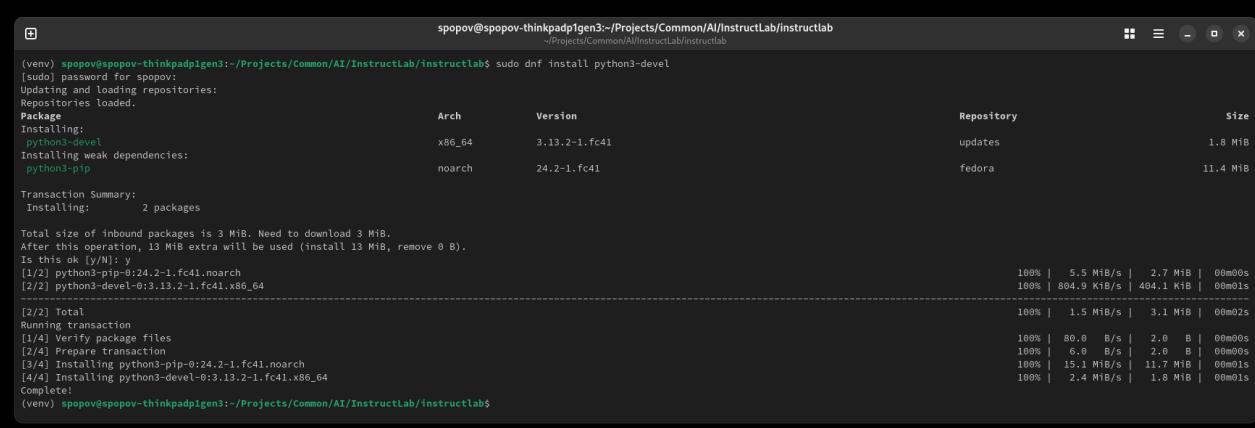
× Encountered error while generating package metadata.
  See above for output.

note: This is an issue with the package mentioned above, not pip.
hint: See above for details.
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

The error message indicates that the build process cannot find the header file "Python.h." This header is part of the Python development package, which is necessary when compiling C extensions for Python. On Fedora, you'll need to install the appropriate development package.

Run the following command to install the Python 3 development libraries:

```
sudo dnf install python3-devel
```



```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ sudo dnf install python3-devel
[sudo] password for spopov:
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository Size
Installing:
python3-devel x86_64 3.13.2-1.fc41 updates 1.8 MiB
Installing weak dependencies:
python3-pip noarch 24.2-1.fc41 fedora 11.4 MiB
Transaction Summary:
Installing: 2 packages
Total size of inbound packages is 3 MiB. Need to download 3 MiB.
After this operation, 13 MiB extra will be used (install 13 MiB, remove 0 B).
Is this ok [y/N]: y
[1/2] python3-pip-0:24.2-1.fc41.noarch 100% | 5.5 MiB/s | 2.7 MiB | 00m00s
[2/2] python3-devel-0:3.13.2-1.fc41.x86_64 100% | 804.9 KiB/s | 404.1 KiB | 00m01s
-----
[2/2] Total 100% | 1.5 MiB/s | 3.1 MiB | 00m02s
Running transaction
[1/4] Verify package files 100% | 80.0 B/s | 2.0 B | 00m00s
[2/4] Prepare transaction 100% | 6.0 B/s | 2.0 B | 00m00s
[3/4] Installing python3-pip-0:24.2-1.fc41.noarch 100% | 15.1 MiB/s | 11.7 MiB | 00m01s
[4/4] Installing python3-devel-0:3.13.2-1.fc41.x86_64 100% | 2.4 MiB/s | 1.8 MiB | 00m01s
Complete!
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

3, Rust Cargo missing

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
~/Projects/Common/AI/InstructLab/instructlab

Downloading safetensors-0.5.1-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.8 kB)
Downloading safetensors-0.5.0-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.8 kB)
INFO: pip is still looking at multiple versions of torchvision to determine which version is compatible with other requirements. This could take a while.
Downloading safetensors-0.4.5-cp313-cp313-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.8 kB)
Downloading safetensors-0.4.3.tar.gz (69 kB)
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... error
error: subprocess-exited-with-error

  Preparing metadata (pyproject.toml) did not run successfully.
  exit code: 1
  10 times of output
  Cargo, the Rust package manager, is not installed or is not on PATH.
  This package requires Rust and Cargo to compile extensions. Install it through
  the system's package manager or via https://rustup.rs/

  Checking for Rust toolchain...
  Read of output

note: This error originates from a subprocess, and is likely not a problem with pip.
error: metadata-generation-failed

Encountered error while generating package metadata.
See above for output.

note: This is an issue with the package mentioned above, not pip.
hint: See above for details.
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

The error indicates that the package you're trying to install requires the Rust toolchain—including Cargo, the Rust package manager—to build some Rust extensions. You'll need to install Rust on your system.

Installing Rust on Fedora You have two common options to install Rust:

Option 1: Using dnf

On Fedora, you can install Rust and Cargo using the package manager by running:

```
sudo dnf install rust cargo
```

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
~/Projects/Common/AI/InstructLab/instructlab

(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ sudo dnf install rust cargo
[sudo] password for spopov:
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository Size
Installing:
cargo x86_64 1.86.0-1.fc41 updates 20.3 MiB
rust x86_64 1.86.0-1.fc41 updates 85.3 MiB
Installing dependencies:
libgit2 x86_64 1.9.0-4.fc41 updates 1.3 MiB
llhttp x86_64 9.2.1-2.fc41 fedora 72.6 KiB
rust-std-static x86_64 1.86.0-1.fc41 updates 153.7 MiB

Transaction Summary:
Installing: 5 packages

Total size of inbound packages is 73 MiB. Need to download 73 MiB.
After this operation, 261 MiB extra will be used (install 261 MiB, remove 0 B).
Is this ok [y/N]: y
[1/5] cargo-0:1.86.0-1.fc41.x86_64 100% | 2.4 MiB/s | 6.5 MiB | 00m03s
[2/5] libgit2-0:1.9.0-4.fc41.x86_64 100% | 3.7 MiB/s | 580.1 KiB | 00m00s
[3/5] llhttp-0:9.2.1-2.fc41.x86_64 100% | 83.8 KiB/s | 32.9 KiB | 00m00s
[4/5] rust-0:1.86.0-1.fc41.x86_64 100% | 5.5 MiB/s | 27.2 MiB | 00m05s
[5/5] rust-std-static-0:1.86.0-1.fc41.x86_64 100% | 7.1 MiB/s | 39.0 MiB | 00m06s
-----
[5/5] Total 100% | 10.9 MiB/s | 73.4 MiB | 00m07s
Running transaction
[1/7] Verify package files 100% | 9.0 B/s | 5.0 B | 00m01s
[2/7] Prepare transaction 100% | 11.0 B/s | 5.0 B | 00m00s
[3/7] Installing rust-std-static-0:1.86.0-1.fc41.x86_64 100% | 203.6 MiB/s | 153.7 MiB | 00m01s
[4/7] Installing rust-0:1.86.0-1.fc41.x86_64 100% | 169.6 MiB/s | 85.3 MiB | 00m01s
[5/7] Installing llhttp-0:9.2.1-2.fc41.x86_64 100% | 10.3 MiB/s | 73.8 KiB | 00m00s
[6/7] Installing libgit2-0:1.9.0-4.fc41.x86_64 100% | 88.8 MiB/s | 1.3 MiB | 00m00s
[7/7] Installing cargo-0:1.86.0-1.fc41.x86_64 100% | 23.3 MiB/s | 20.3 MiB | 00m01s
Complete!
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

1. Missing Cmake


```

spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
~/Projects/Common/AI/InstructLab/instructlab

File "/tmp/pip-build-env-lcxj0i6m/overlay/lib/python3.13/site-packages/setuptools/_distutils/dist.py", line 1021, in run_command
cmd_obj.run()
~~~~~
File "/tmp/pip-build-env-lcxj0i6m/overlay/lib/python3.13/site-packages/setuptools/command/build_ext.py", line 99, in run
_build_ext.run(self)
~~~~~
File "/tmp/pip-build-env-lcxj0i6m/overlay/lib/python3.13/site-packages/setuptools/_distutils/command/build_ext.py", line 368, in run
self.build_extensions()
~~~~~
File "/tmp/pip-build-env-lcxj0i6m/overlay/lib/python3.13/site-packages/setuptools/_distutils/command/build_ext.py", line 484, in build_extensions
self._build_extensions_serial()
~~~~~
File "/tmp/pip-build-env-lcxj0i6m/overlay/lib/python3.13/site-packages/setuptools/_distutils/command/build_ext.py", line 510, in _build_extensions_serial
self.build_extension(ext)
~~~~~
File "<string>", line 87, in build_extension
File "/usr/lib64/python3.13/subprocess.py", line 421, in check_call
raise CalledProcessError(retcode, cmd)
subprocess.CalledProcessError: Command '['./build_bundled.sh', '0.2.0']' returned non-zero exit status 127.
[end of output]

note: This error originates from a subprocess, and is likely not a problem with pip.
ERROR: Failed building wheel for sentencepiece
Successfully built llama-cpp-python numpy rouge-score
Failed to build sentencepiece
ERROR: Failed to build installable wheels for some pyproject.toml based projects (sentencepiece)
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$

```

Sentencepiece’s build process often depends on CMake. On Fedora, you can install CMake by running:

```
sudo dnf install cmake
```

```

spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
~/Projects/Common/AI/InstructLab/instructlab

(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ sudo dnf install cmake
[sudo] password for spopov:
Updating and loading repositories:
Fedora 41 - x86_64 - Updates
Repositories loaded.
Package Arch Version Repository Size
Installing:
cmake x86_64 3.30.8-1.fc41 updates 32.7 MiB
Installing dependencies:
cmake-data noarch 3.30.8-1.fc41 updates 8.2 MiB
cmake-filesystem x86_64 3.30.8-1.fc41 updates 0.0 B
jsoncpp x86_64 1.9.5-8.fc41 fedora 253.4 KiB
rhash x86_64 1.4.4-2.fc41 fedora 349.9 KiB

Transaction Summary:
Installing: 5 packages

Total size of inbound packages is 14 MiB. Need to download 14 MiB.
After this operation, 42 MiB extra will be used (install 42 MiB, remove 0 B).
Is this ok [y/N]: y
[1/5] rhash-0:1.4.4-2.fc41.x86_64 100% | 556.8 KiB/s | 196.0 KiB | 00m00s
[2/5] jsoncpp-0:1.9.5-8.fc41.x86_64 100% | 270.6 KiB/s | 99.3 KiB | 00m00s
[3/5] cmake-filesystem-0:3.30.8-1.fc41.x86_64 100% | 96.6 KiB/s | 16.8 KiB | 00m00s
[4/5] cmake-0:3.30.8-1.fc41.x86_64 100% | 16.6 MiB/s | 10.9 MiB | 00m01s
[5/5] cmake-data-0:3.30.8-1.fc41.noarch 100% | 6.3 MiB/s | 2.3 MiB | 00m00s
-----
[5/5] Total 100% | 7.1 MiB/s | 13.5 MiB | 00m02s
Running transaction
[1/7] Verify package files 100% | 58.0 B/s | 5.0 B | 00m00s
[2/7] Prepare transaction 100% | 12.0 B/s | 5.0 B | 00m00s
[3/7] Installing cmake-filesystem-0:3.30.8-1.fc41.x86_64 100% | 561.6 KiB/s | 7.3 KiB | 00m00s
[4/7] Installing rhash-0:1.4.4-2.fc41.x86_64 100% | 13.9 MiB/s | 355.1 KiB | 00m00s
[5/7] Installing jsoncpp-0:1.9.5-8.fc41.x86_64 100% | 13.8 MiB/s | 254.9 KiB | 00m00s
[6/7] Installing cmake-data-0:3.30.8-1.fc41.noarch 100% | 6.0 MiB/s | 8.8 MiB | 00m01s
[7/7] Installing cmake-0:3.30.8-1.fc41.x86_64 100% | 40.1 MiB/s | 32.7 MiB | 00m01s
Complete!
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ cmake --version
cmake version 3.30.8

CMake suite maintained and supported by Kitware (kitware.com/cmake).
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$

```

4.2. Setting up models

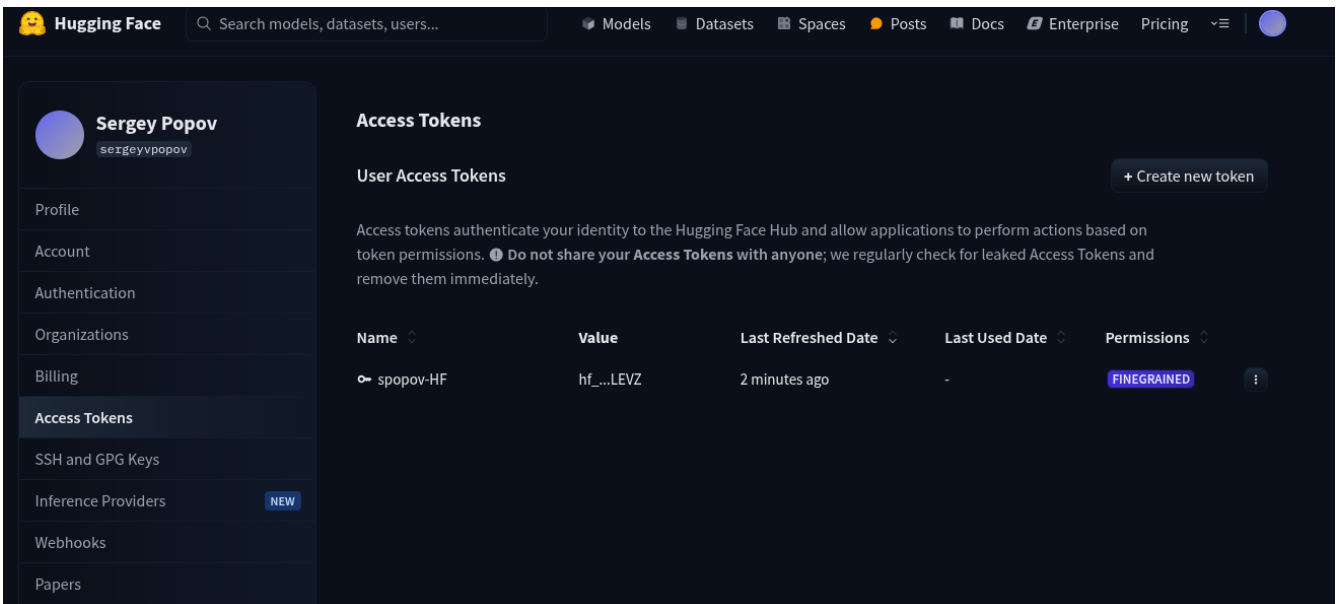
REF: <https://www.redhat.com/en/blog/instructlab-tutorial-installing-and-fine-tuning-your-first-ai-model-part-1>

Install the HF CLI (if not installed):

```
pip install huggingface_hub
```

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab
~/Projects/Common/AI/InstructLab/instructlab
(env) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ pip install huggingface_hub
Requirement already satisfied: huggingface_hub in ./venv/lib64/python3.13/site-packages (0.30.2)
Requirement already satisfied: filelock in ./venv/lib64/python3.13/site-packages (from huggingface_hub) (3.18.0)
Requirement already satisfied: fsspec>=2023.5.0 in ./venv/lib64/python3.13/site-packages (from huggingface_hub) (2024.6.1)
Requirement already satisfied: packaging>=20.9 in ./venv/lib64/python3.13/site-packages (from huggingface_hub) (24.2)
Requirement already satisfied: pyyaml>=5.1 in ./venv/lib64/python3.13/site-packages (from huggingface_hub) (6.0.2)
Requirement already satisfied: requests in ./venv/lib64/python3.13/site-packages (from huggingface_hub) (2.32.3)
Requirement already satisfied: tqdm>=4.42.1 in ./venv/lib64/python3.13/site-packages (from huggingface_hub) (4.67.1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in ./venv/lib64/python3.13/site-packages (from huggingface_hub) (4.13.2)
Requirement already satisfied: charset-normalizer<4,>=2 in ./venv/lib64/python3.13/site-packages (from requests->huggingface_hub) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in ./venv/lib64/python3.13/site-packages (from requests->huggingface_hub) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in ./venv/lib64/python3.13/site-packages (from requests->huggingface_hub) (2.4.0)
Requirement already satisfied: certifi>=2017.4.17 in ./venv/lib64/python3.13/site-packages (from requests->huggingface_hub) (2025.1.31)
(env) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

generate and download Hf token, keep it safe



Login to HF using token:

You'll be prompted to enter your access token, which you can generate from Hugging Face settings. Once logged in, the CLI stores the token in your home directory (typically in ~/.huggingface/hub), and many tools will automatically pick it up.

```
huggingface-cli login
```


Now we have list of commands

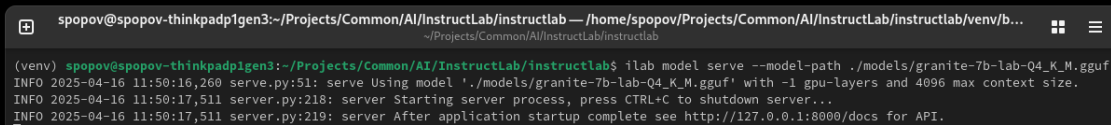
```
Commands:
 chat      Run a chat using the modified model
 convert   Converts model to GGUF
 download  Download the model(s) to train
 serve     Start a local server
 test      Runs basic test to ensure model correctness
 train     Takes synthetic data generated locally with `ilab generate`...
```

4.2.1. Serving the model

At this point, you are able to chat with your model. You will need two different sessions, one for serving the model and the other for chatting with the model.

Begin by starting the model server:

```
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ ls ./models/ -lart
total 3985404
drwxr-xr-x. 1 spopov spopov      22 Apr 16 11:22 .cache
-rw-r--r--. 1 spopov spopov 4081050336 Apr 16 11:23 granite-7b-lab-Q4_K_M.gguf
drwxr-xr-x. 1 spopov spopov      802 Apr 16 11:23 ..
drwxr-xr-x. 1 spopov spopov      64 Apr 16 11:23 .
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ ilab model serve --model-path
./models/granite-7b-lab-Q4_K_M.gguf
```



```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab — /home/spopov/Projects/Common/AI/InstructLab/instructlab/venv/b...
~/Projects/Common/AI/InstructLab/instructlab
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ ilab model serve --model-path ./models/granite-7b-lab-Q4_K_M.gguf
INFO 2025-04-16 11:50:16,260 server.py:51: serve Using model './models/granite-7b-lab-Q4_K_M.gguf' with -1 gpu-layers and 4096 max context size.
INFO 2025-04-16 11:50:17,511 server.py:218: server Starting server process, press CTRL+C to shutdown server...
INFO 2025-04-16 11:50:17,511 server.py:219: server After application startup complete see http://127.0.0.1:8000/docs for API.
```

Server started ! We can communicate with it now

Now start a new session in new terminal window

If you are using upstream InstructLab, start venv:

```
source venv/bin/activate
ilab model chat
```

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ source venv/bin/activate
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$ ilab model chat
Welcome to InstructLab Chat w/ ./MODELS/GRANITE-7B-LAB-Q4_K_M.GGUF (type /h for help)

>>> Good day
Good day to you! How can I assist?
elapsed 2.535 seconds

>>> Thank you. What is the role of Noam Chomsky in creating LLMs?
Noam Chomsky, a renowned linguist and cognitive scientist, has played an influential role in the development of Artificial Intelligence (AI) and Language Learning Models (LLMs). Although he did not directly create LLMs, his theories and ideas have significantly contributed to their foundation.

Chomsky's transformational-generative grammar framework has been instrumental in shaping the way we understand language structure and generation. This framework posits that human language acquisition is a result of innate mental structures, which is an essential concept in AI and LLMs. The idea is that language learning algorithms can be designed to leverage these innate mental structures, enabling machines to better understand and generate human language.

Moreover, Chomsky's work on linguistic relativity, or the Sapir-Whorf Hypothesis, has influenced the development of AI and LLMs. This hypothesis suggests that the structure of a language influences its speakers' cognitive processes and worldview. As a result, AI systems designed to learn and generate language must be exposed to diverse linguistic inputs to ensure comprehensive and accurate understanding and generation.

In summary, while Noam Chomsky did not directly create LLMs, his theories and ideas have significantly contributed to their foundation. His work on transformational-generative grammar and linguistic relativity has helped shape the way we understand language structure and acquisition, ultimately influencing the development of AI and LLMs.

If you have any further questions about Noam Chomsky, AI, or LLMs, please feel free to ask!
elapsed 95.020 seconds

>>>
```

4.2.2. Troubleshooting model installation

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
UF --filename granite-7b-lab-Q4_K_M.gguf
Downloading model from instructlab/granite-7b-lab-GGUF@main to models...
Downloading model failed with the following Hugging Face Hub error: 401 Client Error: Unauthorized for url: https://huggingface.co/api/models/instructlab/granite-7b-lab-GGUF/tree/main?recursive=True&expand=False (Request ID: Root=1-67ff618d-4b7bf4c96ac9c0473589aacb;2b2653cc-34ed-47be-a750-14cda3ab7810)

Invalid credentials in Authorization header
(venv) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/instructlab$
```

This error indicates that your request to download the model is unauthorized because the Hugging Face Hub is expecting valid credentials. In most cases, you'll need to provide your Hugging Face access token so that the system can authenticate you.

4.3. Installing tensorflow

On Fedora, the python3.X-venv package name only exists on Debian/Ubuntu—it doesn't exist in Fedora's repos. The good news is that the venv module is built right into the Fedora packaged interpreter, so you only need:

```
sudo dnf install python3.11 python3.11-devel
```

```

spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow
~Projects/Common/Al/InstructLab/tensorflow

spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ sudo dnf install python3.11 python3.11-devel
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository Size
Installing:
python3.11 x86_64 3.11.12-1.fc41 updates 31.5 KiB
python3.11-devel x86_64 3.11.12-1.fc41 updates 929.5 KiB
Installing dependencies:
python3.11-libs x86_64 3.11.12-1.fc41 updates 43.7 MiB
Installing weak dependencies:
python3.11-tkinter x86_64 3.11.12-1.fc41 updates 2.3 MiB
Transaction Summary:
Installing: 4 packages
Total size of inbound packages is 10 MiB. Need to download 10 MiB.
After this operation, 47 MiB extra will be used (install 47 MiB, remove 0 B).
Is this ok [y/N]: y
[1/4] python3.11-0:3.11.12-1.fc41.x86_64 100% | 878.0 B/s | 26.0 KiB | 00m30s
>>> curl error (28): Timeout was reached for http://mirrors.chroot.ro/fedora/linux/updates/41/Everything/x86_64/Packages/p/python3.11-3.11.12-1.fc41.x86_64.rpm [Failed to connect to mirrors.chroot.ro port 80 after
[2/4] python3.11-devel-0:3.11.12-1.fc41.x86_64 100% | 8.7 KiB/s | 266.4 KiB | 00m31s
>>> curl error (28): Timeout was reached for http://mirrors.chroot.ro/fedora/linux/updates/41/Everything/x86_64/Packages/p/python3.11-devel-3.11.12-1.fc41.x86_64.rpm [Failed to connect to mirrors.chroot.ro port 8
[3/4] python3.11-tkinter-0:3.11.12-1.fc41.x86_64 100% | 1.6 MiB/s | 415.1 KiB | 00m00s
[4/4] python3.11-libs-0:3.11.12-1.fc41.x86_64 100% | 315.3 KiB/s | 9.5 MiB | 00m31s
>>> curl error (28): Timeout was reached for http://mirrors.chroot.ro/fedora/linux/updates/41/Everything/x86_64/Packages/p/python3.11-libs-3.11.12-1.fc41.x86_64.rpm [Failed to connect to mirrors.chroot.ro port 80
-----
[4/4] Total 100% | 336.0 KiB/s | 10.2 MiB | 00m31s
Running transaction
[1/6] Verify package files
[2/6] Prepare transaction
[3/6] Installing python3.11-libs-0:3.11.12-1.fc41.x86_64 100% | 50.0 B/s | 4.0 B | 00m00s
[4/6] Installing python3.11-0:3.11.12-1.fc41.x86_64 100% | 7.0 B/s | 4.0 B | 00m01s
[5/6] Installing python3.11-devel-0:3.11.12-1.fc41.x86_64 100% | 25.0 MiB/s | 44.2 MiB | 00m02s
[6/6] Installing python3.11-tkinter-0:3.11.12-1.fc41.x86_64 100% | 3.2 MiB/s | 32.7 KiB | 00m00s
Complete! 100% | 6.0 MiB/s | 963.2 KiB | 00m00s
100% | 2.2 MiB/s | 2.3 MiB | 00m01s
spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ python3 --version
Python 3.11.2
spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ python3.11 -m venv ~/venvs/tf311
spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ source ~/venvs/tf311/bin/activate
(tf311) spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ python -m ensurepip --upgrade
Looking in links: /tmp/tmpxd143h2q
Requirement already satisfied: setuptools in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (69.2.0)
Requirement already satisfied: pip in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (24.2)
(tf311) spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ pip install --upgrade pip setuptools wheel
Collecting pip
Using cached pip-25.0.1-py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: setuptools in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (69.2.0)
Collecting setuptools
Using cached setuptools-78.1.0-py3-none-any.whl.metadata (6.6 kB)
Collecting wheel
Downloading wheel-0.45.1-py3-none-any.whl.metadata (2.3 kB)
Using cached pip-25.0.1-py3-none-any.whl (1.8 MB)
Using cached setuptools-78.1.0-py3-none-any.whl (1.3 MB)
Downloading wheel-0.45.1-py3-none-any.whl (72 kB)
Installing collected packages: wheel, setuptools, pip
Attempting uninstall: setuptools
Found existing installation: setuptools 69.2.0
Uninstalling setuptools-69.2.0:
Successfully uninstalled setuptools-69.2.0
Attempting uninstall: pip
Found existing installation: pip 24.2
Uninstalling pip-24.2:
Successfully uninstalled pip-24.2
Successfully installed pip-25.0.1 setuptools-78.1.0 wheel-0.45.1
(tf311) spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ pip install tensorflow
Collecting tensorflow
Downloading tensorflow-2.19.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
Collecting absl-py>=1.0.0 (from tensorflow)

```

python3.11 -m venv ~/venvs/tf311

```

Python 3.11.2
spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ python3.11 -m venv ~/venvs/tf311
spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ source ~/venvs/tf311/bin/activate
(tf311) spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ python -m ensurepip --upgrade
Looking in links: /tmp/tmpxd143h2q
Requirement already satisfied: setuptools in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (69.2.0)
Requirement already satisfied: pip in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (24.2)
(tf311) spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ pip install --upgrade pip setuptools wheel
Requirement already satisfied: pip in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (24.2)
Collecting pip
Using cached pip-25.0.1-py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: setuptools in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (69.2.0)
Collecting setuptools
Using cached setuptools-78.1.0-py3-none-any.whl.metadata (6.6 kB)
Collecting wheel
Downloading wheel-0.45.1-py3-none-any.whl.metadata (2.3 kB)
Using cached pip-25.0.1-py3-none-any.whl (1.8 MB)
Using cached setuptools-78.1.0-py3-none-any.whl (1.3 MB)
Downloading wheel-0.45.1-py3-none-any.whl (72 kB)
Installing collected packages: wheel, setuptools, pip
Attempting uninstall: setuptools
Found existing installation: setuptools 69.2.0
Uninstalling setuptools-69.2.0:
Successfully uninstalled setuptools-69.2.0
Attempting uninstall: pip
Found existing installation: pip 24.2
Uninstalling pip-24.2:
Successfully uninstalled pip-24.2
Successfully installed pip-25.0.1 setuptools-78.1.0 wheel-0.45.1
(tf311) spopov@spopov-thinkpad1gen3:~/Projects/Common/Al/InstructLab/tensorflow$ pip install tensorflow
Collecting tensorflow
Downloading tensorflow-2.19.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
Collecting absl-py>=1.0.0 (from tensorflow)

```

Activate the venv

source ~/venvs/tf311/bin/activate

Bootstrap pip if needed If you see an error about missing pip, run:

python -m ensurepip --upgrade

Upgrade installer tooling

```
pip install --upgrade pip setuptools wheel
```

```
spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/tensorflow
~/Projects/Common/AI/InstructLab/tensorflow

(tf311) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/tensorflow$ pip install --upgrade pip setuptools wheel
Requirement already satisfied: pip in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (24.2)
Collecting pip
  Using cached pip-25.0.1-py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: setuptools in /home/spopov/venvs/tf311/lib64/python3.11/site-packages (69.2.0)
Collecting setuptools
  Using cached setuptools-78.1.0-py3-none-any.whl.metadata (6.6 kB)
Collecting wheel
  Downloading wheel-0.45.1-py3-none-any.whl.metadata (2.3 kB)
Using cached pip-25.0.1-py3-none-any.whl (1.8 MB)
Using cached setuptools-78.1.0-py3-none-any.whl (1.3 MB)
Downloading wheel-0.45.1-py3-none-any.whl (72 kB)
Installing collected packages: wheel, setuptools, pip
  Attempting uninstall: setuptools
    Found existing installation: setuptools 69.2.0
    Uninstalling setuptools-69.2.0:
      Successfully uninstalled setuptools-69.2.0
  Attempting uninstall: pip
    Found existing installation: pip 24.2
    Uninstalling pip-24.2:
      Successfully uninstalled pip-24.2
Successfully installed pip-25.0.1 setuptools-78.1.0 wheel-0.45.1
(tf311) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/tensorflow$ pip install tensorflow
Collecting tensorflow
```

Install TensorFlow

For the full package (CPU + GPU support):

```
pip install tensorflow
```

Verify the install

```
python -c "import tensorflow as tf; print(tf.__version__)"
```

```
Successfully installed MarkupSafe-3.0.2 absl-py-2.2.2 astunparse-1.6.3 certifi-2025.1.31 charset-normalizer-3.4.1 flatbuffers-25.2.10 gast-0.6.0 google-pasta-0.2.0 grpcio-1.71.0 h5py-3.13.0 idna-3.10 keras-3.9.2
libclang-18.1.1 markdown-3.8 markdown-it-py-3.0.0 mdurl-0.1.2 ml-dtypes-0.5.1 namex-0.0.8 numpy-2.1.3 opt-einsum-3.4.0 optree-0.15.0 packaging-24.2 protobuf-5.29.4 pygments-2.19.1 requests-2.32.3 rich-14.0.0 six-
1.17.0 tensorboard-2.19.0 tensorboard-data-server-0.7.2 tensorflow-2.19.0 tensorflow-io-gcs-filesystem-0.37.1 termcolor-3.0.1 typing-extensions-4.13.2 urllib3-2.4.0 werkzeug-3.1.3 wrapt-1.17.2
(tf311) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/tensorflow$ python -c "import tensorflow as tf; print(tf.__version__)"
2025-04-18 20:31:53.975419: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:32] Could not find cuda drivers on your machine, GPU will not be used.
2025-04-18 20:31:53.979910: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:32] Could not find cuda drivers on your machine, GPU will not be used.
2025-04-18 20:31:53.989411: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
E0000 00:00:1745001114.006196 578537 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been registered
E0000 00:00:1745001114.011124 578537 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already been registered
W0000 00:00:1745001114.025935 578537 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1745001114.025956 578537 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1745001114.025959 578537 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1745001114.025963 578537 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
2025-04-18 20:31:54.031137: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
2.19.0
(tf311) spopov@spopov-thinkpadp1gen3:~/Projects/Common/AI/InstructLab/tensorflow$
```

(Optional) Making Python 3.11 your default python3

If you really want python3 to point at Python 3.11, you can use Fedora's alternatives system:

```
sudo alternatives --install /usr/bin/python3 python3 /usr/bin/python3.11 2
```

Then choose which version is the default:

```
sudo alternatives --config python3
```

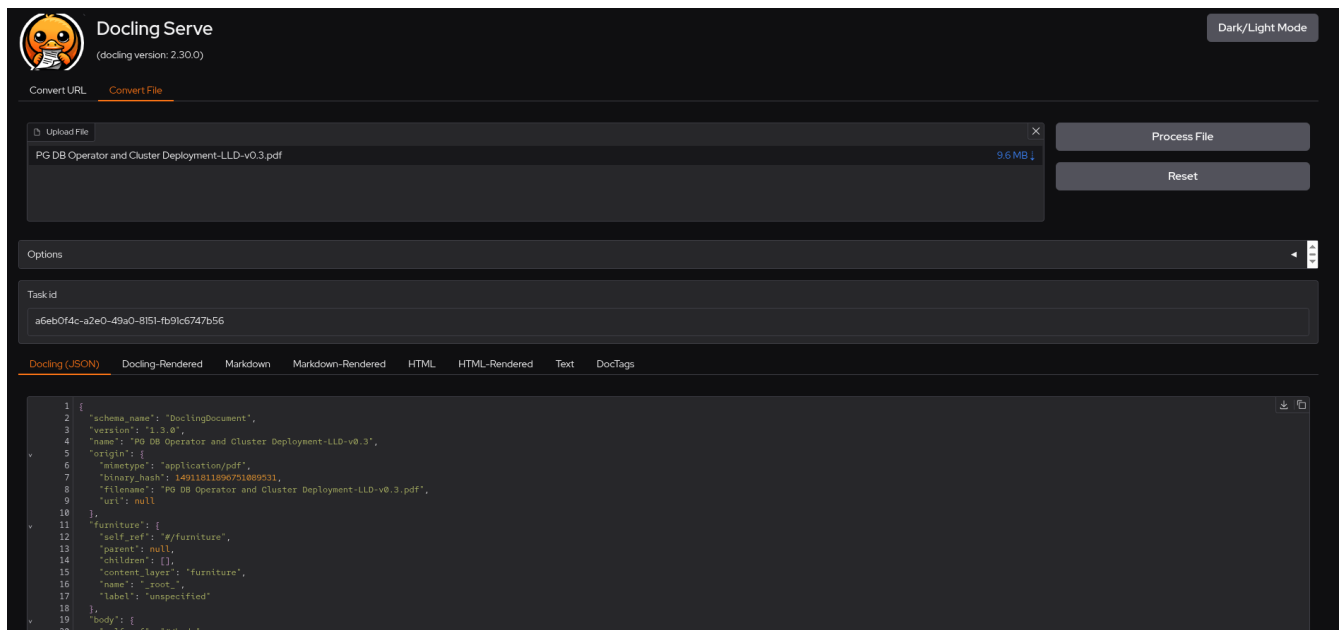
—but be cautious, as changing the system python3 can affect other tools.

Chapter 5. Model Training

5.1. Docling file conversion

```
spopov@spopov-thinkpadp1gen3:~/Projects/Core42/VPN — podman run -p 5001:5001 -e DOCLING_SERVE_ENABLE_UI=true quay.io/docling-project/docling-serve
~/Projects/Core42/VPN
spopov@spopov-thinkpadp1gen3:~/Projects/Core42/VPN — podman run -p 5001:5001 -e DOCLING

INFO: 127.0.0.1:45510 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:45518 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:51960 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:51964 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:36370 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:36374 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:32822 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:32832 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:38644 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:38656 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:41594 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:41596 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:42406 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:42416 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:57416 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:57432 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:44568 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:44572 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:41616 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:41620 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:44496 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:44504 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60422 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:60426 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:44400 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:54950 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:54952 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:58510 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53058 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:53064 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:51168 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:51184 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:41958 - "GET /v1alpha/status/poll/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56?wait=5 HTTP/1.1" 200 OK
INFO: 127.0.0.1:41960 - "GET /v1alpha/result/a6eb0f4c-a2e0-49a0-8151-fb91c6747b56 HTTP/1.1" 200 OK
INFO: 172.20.7.6:43732 - "GET /ui/assets/index-BXdxtn7e.js HTTP/1.1" 200 OK
INFO: 172.20.7.6:43730 - "GET /ui/assets/index-C7AHfj1r.js HTTP/1.1" 200 OK
INFO: 172.20.7.6:43734 - "GET /ui/assets/index-8qbykVEb.js HTTP/1.1" 200 OK
INFO: 172.20.7.6:43752 - "GET /ui/assets/index-C8bPzdeA.js HTTP/1.1" 200 OK
INFO: 172.20.7.6:43748 - "GET /ui/assets/index-BWtcSLEP.js HTTP/1.1" 200 OK
INFO: 172.20.7.6:43734 - "GET /ui/assets/frontmatter-DSPTV04n.js HTTP/1.1" 200 OK
INFO: 172.20.7.6:43732 - "GET /ui/assets/yaml-DsCXHVtH.js HTTP/1.1" 200 OK
INFO: 172.20.7.6:43732 - "GET /ui/assets/index-C7bgy9xM.js HTTP/1.1" 200 OK
INFO: 172.20.7.6:43732 - "GET /ui/ HTTP/1.1" 200 OK
```



Chapter 6. OCP Operation manual

Unresolved directive in README.adoc -
include::content/OCP/110_manual/005_ops_manual.adoc[leveloffset=+2]

RHO AI

Appendix A: RH/OCP References

1. DG Operator Guide: https://access.redhat.com/documentation/en-us/red_hat_data_grid/8.3/html-single/data_grid_operator_guide/index
2. DG SSO on OCP: <https://developers.redhat.com/blog/2021/04/23/integrate-red-hat-data-grid-and-red-hats-single-sign-on-technology-on-red-hat-openshift>
3. DG Services: <https://developers.redhat.com/articles/2021/06/08/create-and-manage-red-hat-data-grid-services-hybrid-cloud>
4. DG Operator: <https://developers.redhat.com/articles/2024/01/17/how-install-and-upgrade-data-grid-8-operator>
5. AMQ HA on OCP: https://access.redhat.com/documentation/en-us/red_hat_amq/7.6/html/deploying_amq_broker_on_openshift/high-availability-for-broker-on-ocp_broker-ocp